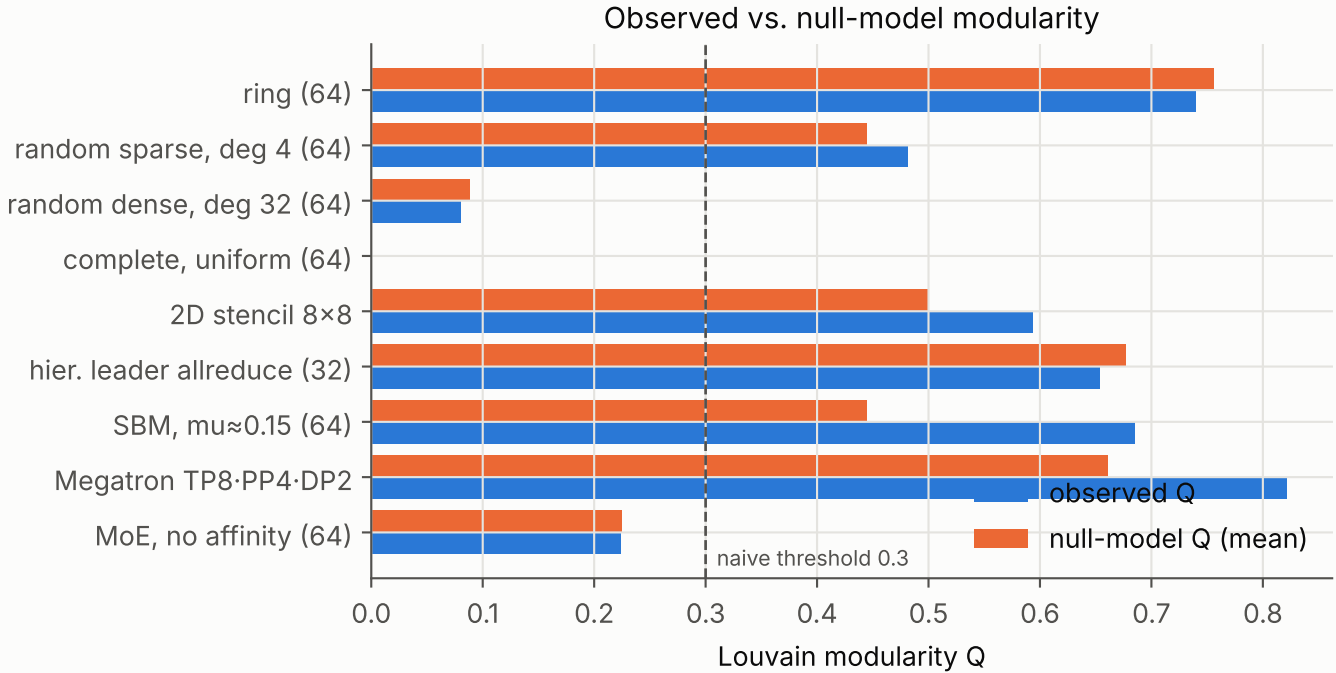
Observed vs. null-model modularity
ring (64)
random sparse, deg 4 (64)
random dense, deg 32 (64)
complete, uniform (64)
2D stencil 8×8
hier. leader allreduce (32)
SBM, mu≈0.15 (64)
Megatron TP8·PP4·DP2
MoE, no affinity (64)
observed Q
null-model Q (mean)
naive threshold 0.3
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Louvain modularity Q